

3. [15 marks]

2.11 (2014:40)

Citric acid is the active ingredient in some bathroom and kitchen cleaning solutions. A student determined the content of citric acid in a cleaner by titration with sodium hydroxide solution.

The sodium hydroxide solution first needed to be standardised. To do this, the student dissolved approximately 4 g of sodium hydroxide pellets in water, to give an approximately 0.1 mol L^{-1} solution. This solution was standardised by titrating 20.00 mL of the NaOH solution with a 0.105 mol L^{-1} standard hydrochloric acid solution. The average titration volume was 17.45 mL.

- (a) Explain why sodium hydroxide is not suitable as a primary standard. [2]

- (b) Show that the concentration of the sodium hydroxide solution is $0.0916 \text{ mol L}^{-1}$. Show sufficient workings to justify your answer. [3]

The student then weighed a 10.00 mL aliquot of the cleaner and found it weighed 10.4 g. This 10.00 mL aliquot was next diluted to 100.0 mL in a volumetric flask. Against the standardised sodium hydroxide solution, 20.00 mL aliquots of the diluted cleaner were titrated. The table below shows the results of the titrations.

Titre	1	2	3	4
Final reading (mL)	25.30	23.55	22.40	22.25
Initial reading (mL)	3.50	2.70	1.50	1.30
Titre volume (mL)				

- (c) Calculate the average titre volume to be used in the calculation of the citric acid content. [2]

- (d) Given that citric acid ($\text{C}_6\text{H}_8\text{O}_7$) is a weak triprotic acid, determine the percentage composition by mass of citric acid in the cleaner. The molar mass of citric acid is $192.124 \text{ g mol}^{-1}$. [6]

[illegible]

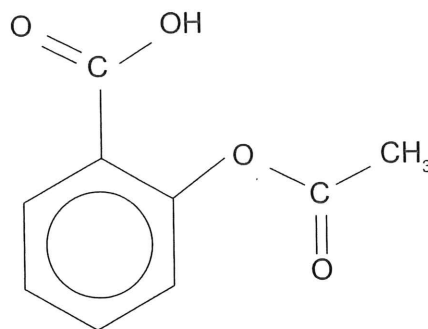
- (e) Select a suitable indicator for this titration from the table below. Explain your choice. [2]

Indicator	Colour change (low pH - high pH)	pH
Methyl yellow	red-yellow	2.4-4.0
Litmus	red-blue	5.0-8.0
Bromothymol blue	yellow-blue	6.0-7.6
Thymol blue	yellow-blue	8.0-9.6

4. [20 marks]

2.11 (2015:36)

Aspirin is one of the most popular and readily available pain-relieving drugs. The structure of aspirin is given below:



Aspirin contains two functional groups.

- (a) (i) On the diagram of aspirin above, circle the **two** functional groups. Label them **A** and **B**. [2]

- (ii) Name each functional group. [2]

Functional group A: _____

Functional group B: _____

- (b) In one commercial brand of aspirin, each '300 mg tablet' is claimed to contain 100% aspirin. To determine the actual percentage by mass of aspirin in an aspirin tablet, the following procedure, involving a back titration, was used.

Step 1: Three aspirin tablets, each with a mass of 300.0 mg, were crushed and dissolved in excess sodium hydroxide solution. Exactly 100.0 mL of 0.204 mol L⁻¹ solution of sodium hydroxide was used. The mixture was boiled to ensure complete reaction.

Step 2: The excess sodium hydroxide solution was titrated with hydrochloric acid as follows: 20.0 mL of the solution from step 1 was pipetted into a conical flask and 0.125 mol L⁻¹ hydrochloric acid was placed in the burette. The indicator, phenolphthalein, was used and an average titre of 17.89 mL of hydrochloric acid was required to reach the end-point.

Notes:

- Assume that any other chemicals present in an aspirin tablet are inert and will not react with either NaOH(aq) or HCl(aq).
- Phenolphthalein is colourless at a pH less than 8.3 and pink at a pH greater than 10.0.

- (i) This is a titration between a strong acid and strong base. Strong acid–strong base titrations typically result in an equivalence point with a pH close to 7. Phenolphthalein was chosen as the indicator for this titration. Considering all of the species present in the solution at the equivalence point, explain why phenolphthalein is a suitable indicator to show the end-point. Support your answer with a suitable equation. [3]

- (ii) Calculate how many moles of hydroxide ions reacted with the aspirin. [5]

- (iii) Each aspirin molecule requires two hydroxide ions for complete reaction. Calculate the percentage by mass of aspirin in one aspirin tablet. (The molar mass of aspirin is $180.154 \text{ g mol}^{-1}$.) [4]

An important procedure in volumetric analysis is the washing of equipment with the appropriate solution prior to the titration in order to minimise experimental error.

- (c) Before performing the experiment, the glassware was washed with the solutions given in the table. Complete the table below by stating the effect of the washing. [4]

Washing procedure	Effect on the volume of hydrochloric acid used	Effect on the % of aspirin calculated
The conical flask was washed with distilled water.		
The burette was washed with distilled water.		

The mole ratio of HCl to HNO_3 required is 3 mol HCl to 1 mol HNO_3 . The quantitative analysis confirms the ratio as correct.

(Note: If the unrounded values are used, you will get a ratio of 3.0318 HCl to 1.000 HNO_3 . This is still acceptable within the limitations of experimental errors.)

3. (2014:40) This problem requires the determination of the content of a sample of citric acid applying volumetric analysis calculations.

a) Explanation why sodium hydroxide cannot be used as a primary standard and has to be standardised prior to using it.

A good primary standard is a chemical that should be available in a pure form with a precise formula. It should not react with the moisture or carbon dioxide in the air and should have a relatively high molar mass to ensure a greater degree of accuracy in weighing. Sodium hydroxide has a low formula mass (39.998). It is not obtained pure as it is highly reactive. As solid pellets, NaOH gradually absorbs water from the air and become a liquid (deliquescent). It also loses its purity as it reacts with carbon dioxide in the air to gradually form Na_2CO_3 . If NaOH is to be used as a standard solution for any titration, it should be standardised first using a solution that is either a primary standard or one that has been standardised very recently, such as HCl .

b) To show that the concentration of the solution used in this calculation is $0.0916 \text{ mol L}^{-1}$.

$m(\text{NaOH})$ pellets taken = about 4 g.

Approximate concentration of the solution made: 0.1 mol L^{-1} .

The reaction with HCl is: $\text{NaOH} + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O}(\ell)$

$$n(\text{HCl}) = c \times v = 0.105 \text{ mol L}^{-1} \times 0.01745 \text{ L} = 1.83225 \times 10^{-3} \text{ mol}$$

$$n(\text{NaOH}) = n(\text{HCl}) = 1.83225 \times 10^{-3} \text{ mol}$$

$$\text{Therefore, } c[\text{NaOH}] = (n \div v) = (1.83225 \times 10^{-3} \text{ mol} \div 0.020 \text{ L}) = 0.0916 \text{ mol L}^{-1} = 9.16 \times 10^{-2} \text{ mol L}^{-1}.$$

c) Calculation of the average titre of NaOH solution:

Final volume (mL)	25.30	23.55	22.40	22.25
Initial volume (mL)	3.50	2.70	1.50	1.30
Titre (mL)	21.80	20.85	20.90	20.95

Discarding the first titre of 21.80 mL (an outlier), the average titre of $\text{NaOH} = 20.90 \text{ mL}$

d) Determining the percentage composition of citric acid by mass.

Volume of the cleaner liquid taken = 10.0 mL

Volume of the stock solution made by dilution = 100.0 mL

Volume of the aliquot used for titration = 20.0 mL

Citric acid is a triprotic acid. This means that one mole of citric acid requires 3 mols of NaOH for complete neutralisation.

$$n(\text{NaOH}) = (c \times v) = (0.0916 \text{ mol L}^{-1} \times 0.02090 \text{ L}) = 1.91444 \times 10^{-3} \text{ mol}$$

$$\text{Therefore } n(\text{citric acid in 20.0 ml dilute}) = (1.91444 \times 10^{-3} \div 3) = 6.3815 \times 10^{-4} \text{ mol}$$

Therefore, 100.0 mL dilute stock solution contains

$$= [(6.3815 \times 10^{-4} \div 20) \times 100] = 3.19073 \times 10^{-3} \text{ mol of citric acid.}$$

$$\text{Therefore, the mass of citric acid in 10.0 mL of the original solution} = (n \times M) = 3.19073 \times 10^{-3} \times 192.184 = 0.6130 \text{ g}$$

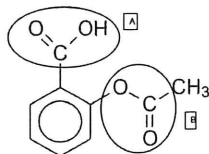
Therefore, the mass of citric acid in the original undiluted solution is 0.6130 g.

Since the mass of the 10.0 mL of the original solution is 10.4 g,

$$\% \text{ by mass of citric acid in the cleaner} = [(0.6130 \div 10.4) \times 100] = 5.89\%$$

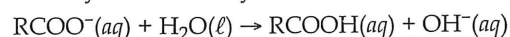
e) Since citric acid is a weak acid and NaOH is a strong base, the equivalence point occurs on the basic side of the pH scale which is between 8 and 10. The hydrolysis of citrate ion will produce OH^- ions according to the equation, $\text{C}_6\text{H}_5\text{O}_7^{3-}(\text{aq}) + 3 \text{H}_2\text{O}(\ell) \rightarrow \text{C}_6\text{H}_8\text{O}_7(\text{aq}) + 3 \text{OH}^-(\text{aq})$. The most suitable indicator from the given list is 'thymol blue' (pH = 8.00 to 9.6).

4. (2015:36) a) i)



ii) Group A: carboxylic acid, Group B: ester

b) i) At the equivalence point the Na^+ and Cl^- ions are neutral. The carboxylate ions react with water to form the carboxylic acid and hydroxide ions



The basic solution means that it is necessary to use an indicator that changes colour higher than 7.

$$\text{ii) } n(\text{H}^+) = cv = 0.125 \times 0.01789 = 2.236 \times 10^{-3} \text{ mol}$$

$$n(\text{OH}^-)_{\text{excess in 20 mL}} = n(\text{H}^+) = 2.236 \times 10^{-3} \text{ mol}$$

$$n(\text{OH}^-)_{\text{excess in 100 mL}} = 2.236 \times 10^{-3} \times 5 = 1.118 \times 10^{-2} \text{ mol}$$

$$n(\text{OH}^-)_{\text{initially in 100 mL}} = cv = 0.204 \times 0.1 = 0.0204 \text{ mol}$$

$$n(\text{OH}^-)_{\text{reacting with aspirin}} = 0.0204 - 0.01118 = 0.00921 \text{ or } 9.21 \times 10^{-3} \text{ mol}$$

$$\text{iii) } n(\text{aspirin}) = n(\text{OH}^-)_{\text{reacting}} = 9.21 \times 10^{-3} \text{ mol}$$

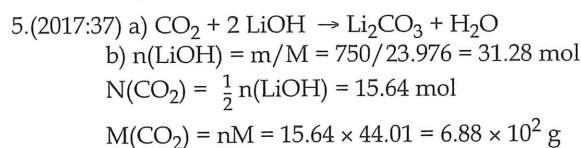
$$m(\text{aspirin}) = 9.21 \times 10^{-3} \times 180.154 = 0.8304 \text{ g}$$

$$m_{(\text{aspirin})} \text{ in 1 tablet} = 0.8304 \div 3 = 0.2768 \text{ g}$$

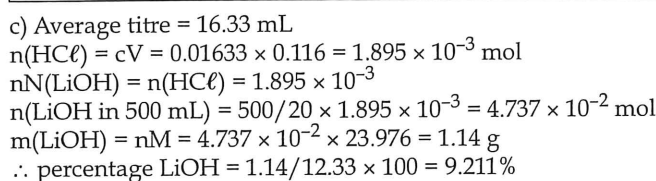
$$\% \text{ aspirin in 1 tablet} = 0.2768 / 0.3 \times 100 = 92.3\%$$

c)

Description		
Washing procedure	Effect on the volume of hydrochloric acid used	Effect on the % of aspirin calculated
The conical flask was washed with distilled water	No effect	No effect
The burette was washed with distilled water	Increase	Decrease



Volume (mL)	1	2	3	4
Final Volume	18.55	34.90	18.50	34.85
Initial Volume	1.50	18.55	2.20	18.50
Titre	17.05	16.35	16.30	16.35

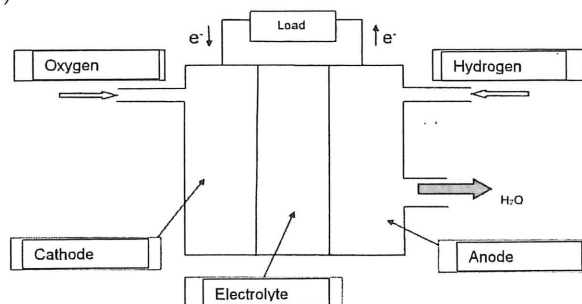


d) Indicator one: Bromothymol blue Indicator two: Phenolphthalein

Explanation: As LiOH is a strong base and HCl is a strong acid, at equivalence they will have a pH of approximately seven because the resulting ions do not undergo hydrolysis to form weak acids or weak bases. The change in pH at equivalence will be large and the vertical part of the pH graph will be many pH units. Any indicator with a pH transition between about 4 and 10 will have appropriate end points for use in this titration.

e) Fuel cells can include CH_4/O_2 and they can even use chemicals such as chlorine. They are not necessarily safe nor always have environmentally friendly products – only the hydrogen oxygen cell does. So the answer must be 'highly efficient' and 'produce constant voltage as the concentration (partial pressure) of the reactants is constant'.

f)



g) The electrolyte completes the circuit and maintains charge balance

h) 1.23 volts

i) The above figure is only true at standard conditions (101 kPa and 25°C). The space vehicle might have been at a lower pressure or lower temperature.

Chapter 14A: Calculations: Organic

1.(2009:03) a) Calculating the empirical formula:

mass of the sample = 9.76 g; mass of carbon dioxide = 17.9 g; mass of water = 4.88 g.

$$m(\text{C}) = [17.9 \times (12.01 \div 44.01)] = 4.88 \text{ g.}$$

$$m(\text{H}) = [4.88 \times (2.016 \div 18.016)] = 0.546 \text{ g.}$$

$$\text{Therefore } m(\text{O}) = [9.76 - (4.88 + 0.546)] = 4.33 \text{ g.}$$

	C	H	O
Mass ratio (g)	4.88	0.546	4.38
Mole ratio	$4.88 \div 12.01$	$0.546 \div 1.008$	$4.33 \div 16.00$
	0.406	0.542	0.271
Simple ratio	$0.406 \div 0.271$	$0.542 \div 0.271$	$0.271 \div 0.271$
Simple ratio	1.5	2.00	1.00
Empirical ratio	3	4	2

Therefore the empirical formula is $\text{C}_3\text{H}_4\text{O}_2$